

Animal Intake Form

Milestone 4

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The artifact, a web application developed in CS340, Client Server Development, connects to a local MongoDB instance to display rescue dog data using the model-view-controller design pattern. It comprises three layers: a database layer, an API layer, and a front-end layer.

I chose this project because it involved a database layer. My previous enhancement dockerized the application stack, making local development for this feature easy. However, the project was read-only, allowing only data viewing without updates. Since it originated from an SNHU lab and lacked the initial dataset, the database was initially empty unless I manually generated a new dataset. This made the animal intake form the most pressing new feature, demonstrating my ability to work with databases.

This enhancement showcases full-stack development skills, particularly the integration of front-end forms with a back-end database. It also demonstrates my ability to create dynamic, user-driven applications. It successfully completed the second and third course outcomes.

I implemented the enhancement by extending the user interface implemented in the dash framework. Since I was not familiar with this framework, it was my first challenge. The next issue I encountered was that the dash framework does not allow multiple callbacks to have the same output. This meant that my newly created callback function to refresh the table in the user interface with new records conflicted with existing callbacks to filter the table. To resolve this, I refactored the code to use a single large callback for all functions that update the data table in the user interface. I also made the other user interface components refresh when the table was changed, ensuring that the map and the pie chart were refreshed automatically without additional work. This best positions the page functionality for future enhancements.